Predicting Fake Job Postings Using Text + Metadata Patterns

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***Abstract*—Job posting sites have made the hiring method easier and more effective but they have also led towards a spike in fake job postings that reduce the security in online job markets. This project analyzes the Employment fraud detector Dataset(EMSCAD) to investigate the Machine learning fraud detection of postings. There are used four ML model- Random Forest, Logistic Regression, SVM,KNN and RNN(LSTM) that includes structural metadata such as salary, Company Profile with TF-IDF text features are put together.**

***Index Terms*—fake job postings, Machine learning, Random Forest, Logistic Regression, SVM, KNN , RNN(LSTM)**

I. Introduction

While, job searching has been easier to find because of the significant of job posting sites such as Glassdoor, Indeed , LinkedIn the amount of fake job postings also has grown [10]. These frauds induce huge amount of economic losses generally by employing theft of identities, gathering personal information and charge fake payments [1].

ML techniques integrating NLP and ML are significant as manual detection of postings is not efficient. But, fake job detection is sometimes challenging since fraudulent postings close to actual ones despite of hidden slight changes in text and form detect the fake job is complicated.

Using the EMSCAD dataset, this project explores classifying the posting job as real or fake.

The main impacts are:

(i) The dataset exploratory type of data analysis methods.

(ii) A feature engineering pipeline that includes metadata reports using TF-IDF text features.

(iii) Contrast with multiple ML models.

(iv)For textual and structures based data a LSTM approach is used.

(v) Evaluate the significant features for interpretation.

II. Dataset Description

For this project used the Employment Scam Aegean Dataset (EMSCAD), that originally collected and published by [1]. Moreover, the dataset is available publicly in kaggle and the filename is `fake_job_postings.csv`.

Dataset contains 18000 job descriptions and from this 800 label as fake. It has both text and structured based data(metadata). moreover, target variable is 0 for real job posting and 1 for fake job posting.

TABLE I
Feature Description Table of the Dataset

Feature Name	Type	Description
text	Text	Text data like- title of job, job description,and company background
has_salary	Binary	salary details is present (1-Yes, 0-No)
has_company	Binary	company profile information is available (1-Yes, 0-No)
fraudulent	Target	job (0-Real, 1-Fake)

TABLE II
Dataset Summary

Attribute	Value
Total Data	18000
Training Set	14304
Test Set	3576
Fake Job Posts	866
Real Job Posts	17014
Target Variable	Fraudulent (0 = Real, 1 = Fake)

III. Literature Review

Research on detect the fake job posting is a online fraud detection and its main goal is to detect the fake job postings by using text analysis methods and ML.

Related Work Analysis:

Automatic Detection of Online Recruit-ment Frauds According to the authors, as usual metadata helpful but not enough for a full solution. Moreover, its need to include more relevant signal like user conduct, network data etc [1] .

Fake Job Recruitment Detection Using Machine Learning Approach This paper shows the EMSCAD dataset and ensemble ML model , RF achieve the maximum accuracy to detect the fake job posts [2].

Fraud-BERT: transformer based context aware online recruitment fraud detection By easily collecting relevant information and hadle the imbalanced data better than normal ML.Moreover, a simplified BERT based transformer obtained the high level performance in fake job detection [3].

Detecting Fake Job Postings Using Bidirectional LSTM highlighted the BI-LSTM that model can detect the fake job postings efficiently and learning the contextual meaning of job descriptions and obtained high level classification accuracy [4].

Detecting Fake Job Posting Using ML Classifications and Ensemble Model. This paper describe the ensemble

TABLE III
RELATED WORK

Paper	Dataset	Methods	Key Finding
Automatic Detection of Online Recruitment Frauds [1]	EMSCAD	Text + Metadata Features (RF, NB, etc.)	Random Forest achieved strong baseline performance.
Fake Job Recruitment Detection Using Machine Learning Approach [2]	Kaggle	Supervised ML + Ensemble Learning	Ensemble models improved performance and handled class imbalance effectively.
Fraud-BERT: Transformer Based Context-Aware Online Recruitment Fraud Detection [3]	EMSCAD	Fine-tuned BERT (Transformer)	BERT achieved very high accuracy for fraud classification.
Detecting Fake Job Postings Using Bidirectional LSTM [4]	Kaggle	Bi-LSTM Deep Learning Model	Captures contextual meaning of job descriptions effectively.
Detecting Fake Job Posting Using ML Classifications and Ensemble Model [5]	EMSCAD	XGBoost + Ensemble Methods	Ensemble models provide robust and stable performance.

learning algorithm. RF, XGBoost gives high level accuracy and strong performance for detect the fake job posting than as usual ML model [5].

A. Exploratory Data Analysis (EDA)

1) The distribution of Dataset: There are a lot of better job post in the dataset compared to fraudulent, which makes it highly unbalanced. This imbalanced needs to be properly evaluated using such as F1-score, ROC-AUC since it could impact model performance.

2) Text Length: As per text length distribution, most of the job listing are set of range. various length pattern are appears in fraudulent post, that might be used as a helpful predictor.

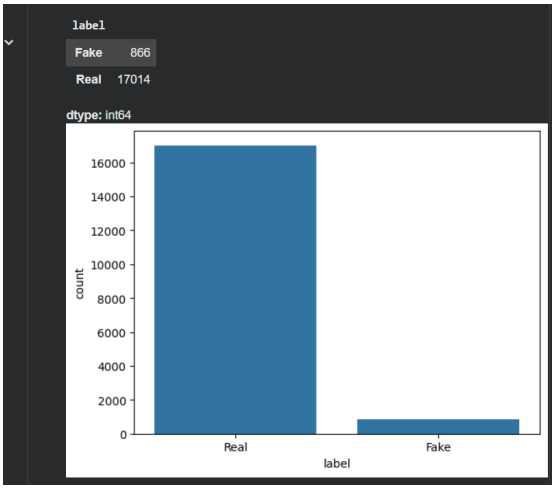


Fig. 1. Real Vs Fake Graph

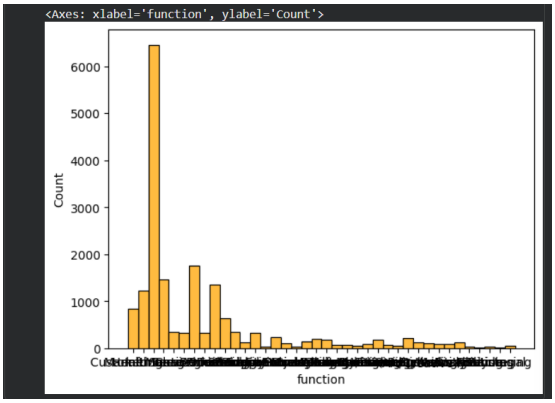


Fig. 2. Text Length

3) Missing Values: Missing values evaluations suggests that incomplete data is present in a set of metadata field. Here, missing company information is present in fake job posting.

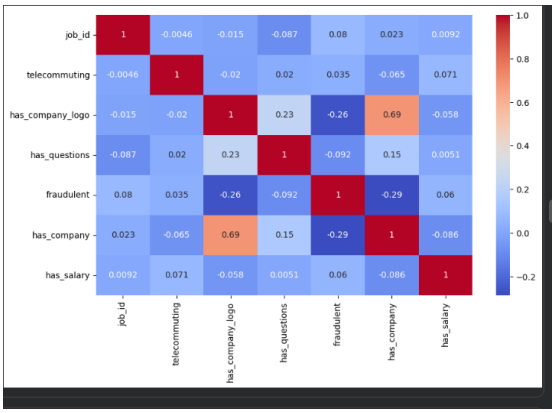


Fig. 3. Missing Values

4) Metadata Pattern: The fake ratio is greater for postings that don't have any company profile than for those postings with one. So, we can say that, most of the fake job posting

company has no profile.

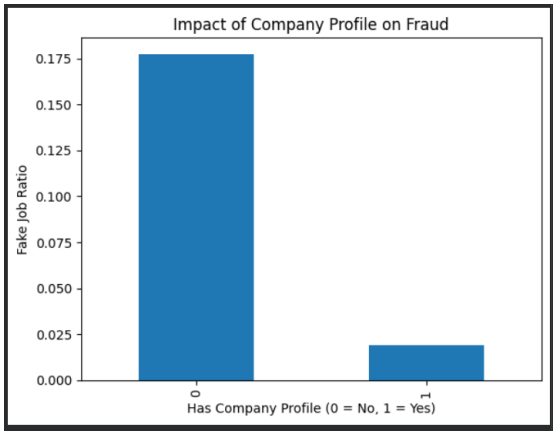


Fig. 4. Metadata Pattern

5) K-means: By using K-means clustering hidden patterns are detected. This indicates the difference between fake and real.

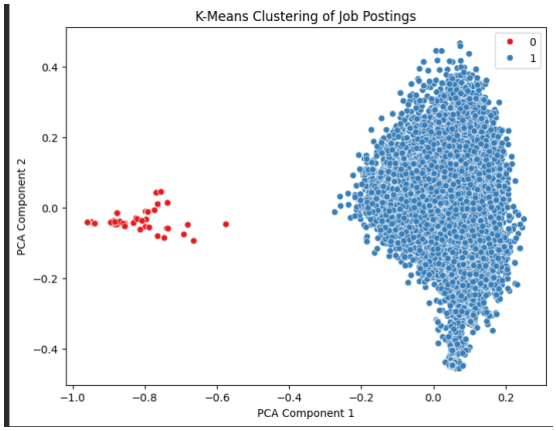


Fig. 5. K-means clustering

IV. Methodology

A. Overview

The experimental pipeline is-

RAW CSV data → EDA → preprocessing → feature Design → model train and evaluation → Interpretations.

B. Data Preprocessing

In order to clean the dataset, missing values handles, unused columns were removes, textual data merged with a feature. Moreover, the data was generated for ML and Deep learning model by using text normalization and tokenization.

C. Feature Designing

TF-IDF and pattern encoding techniques for LSTM were applied to convert a text data. To define the functional patterns different metadata features were collected such as- company information and finding the salary range.

1) Multi-Modal Feature: Textual and metadata features were add to build a multi-modal layout. The method can learn both language pattern and structural errors in job posting. That's why we use "from scipy.sparse import hstack" library to joint the text features with metadata and enable the multi-modal representation for train the model.

D. Train–Test

The dataset is 80% trained and 20% test set .

E. Machine Learning Models

1) Logistic Regression: TF-IDF and metadata are used in base-line model to to detect the job post is real or fake.

2) Random Forest: An ensemble tree based method that gives the information about the relevant features and detects non-linear connections [6].

3) Support Vector Machine : SVM useful for high level sparse features of TF-IDF [7].

4) KNN: KNN used for similarities found in the integrated feature for defining the postings.

F. LSTM

Continious interactions in job descriptions were identified the LSTM based RNN. using metadata features model is significantly improved. We use 3 epoch, 32 batch size and 0.1 validation split.

G. Evaluation Metrics

Accuracy, Precision, Recall,F1-score, ROC - AUC used to check the model outcomes in order to perform a complete check especially for the imbalance class.

V. Results and Analysis

A. Performance Comparison

TABLE IV
Performance of Models on Fake Job Detection

Model	Accuracy	Precision (Fake)	Recall (Fake)	F1-score (Fak
Logistic Regression	0.97	1.00	0.32	0.49
Random Forest	0.98	0.99	0.57	0.72
SVM	0.97	0.96	0.52	0.68
KNN	0.97	0.80	0.59	0.68

Random Forest performance is better than other method because in the table we see that it has highest F1-score and accuracy which shows the best balance between fake precision and recall. It's handling imbalance classes also detect the fake job post than Logistic Regression that has limited recall , SVM, KNN which produce low quality results.

B. Confusion Matrix Inspection

Confusion matrix for Random forest highlights that some fake postings were misclassified since they looked like real job patterns, most of the real job postings were clearly determined.

In the graph , 3394 is true positives real job, 103 is true negatives as fake, 78 false positives fake job but model label this incorrectly as real job posting, 1 false negatives as real job but the model label this incorrectly as fake job posting.

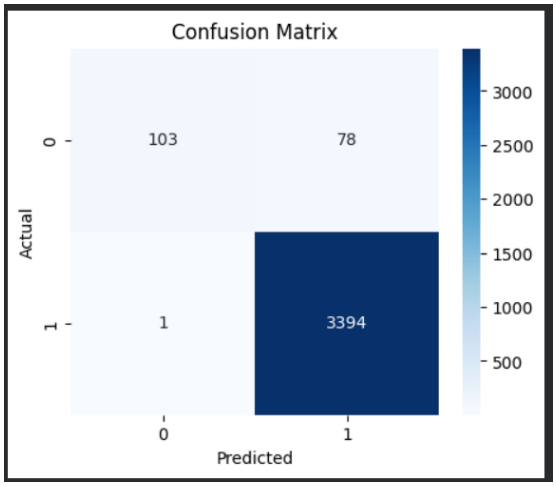


Fig. 6. Confusion Matrix for Random Forest

C. ROC Curve and AUC

Roc showed the strong model classification performance whereas the Auc values detect the accurate separation between real and fake job posting.

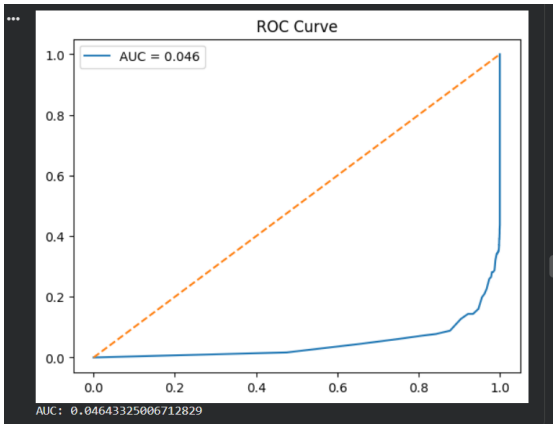


Fig. 7. Roc and Auc

D. Roc and Auc Graph Comparison table

TABLE V
ROC-AUC COMPARISON

Model	AUC Value	Performance Interpretation
KNN	0.111	Highest but bad than random
SVM	0.055	Extremely poor
General ROC	0.046	Lowest

From this table we can say that, KNN ROC curve where AUC value is 0.111 is below 0.5 that indicate that the model perform worse than random error maybe the result of class imbalance.

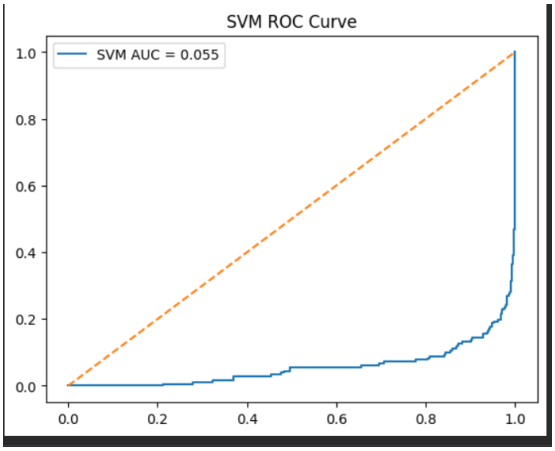


Fig. 8. SVM Roc and Auc

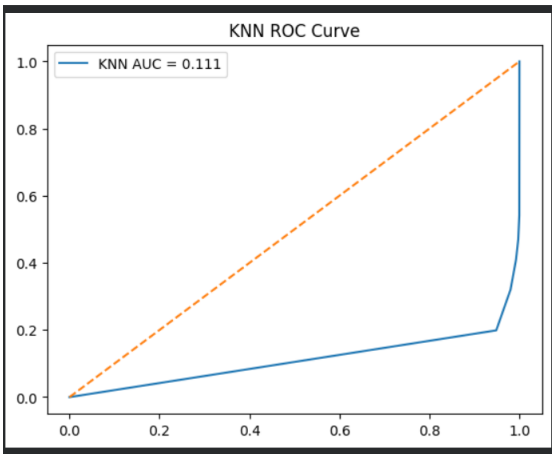


Fig. 9. KNN Roc and Auc

E. Top 10 Feature Analysis

ext based features define the classification due to features impact analysis but metadata features provide extra helping information.

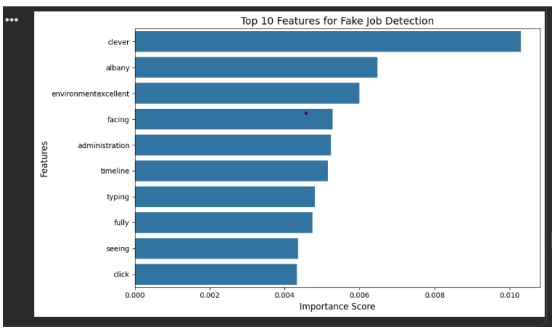


Fig. 10. Top 10 Feature Graph

In the graph we see that, ”clever” is the most important which has highest fraud score. ”Click”,”typing”, ”facing ”

means model is correctly detecting the cmmon scam words ”work from home” or maybe ”data entry”.

Moreover, the use of ”Albany” highlights how significant the spatial metadata to detecting the real one from the fake job postings.

F. K-Means Clustering Analysis

From K-Means Clustering Graph, red 0 is fake and the blue 1 is real. blue represent that majority class is real and red small amount of is fake. thats why random forest achieved high accuarcy score.

G. LSTM Training

Effective training is indicated by the train and validation curve that represent the decline over 3 epoch and control the overfitting is urged by minimum variance.

Test Accuracy:0.9678411483764648. it’s gained 96.78% test accuracy . it’s high since class imbalanced.

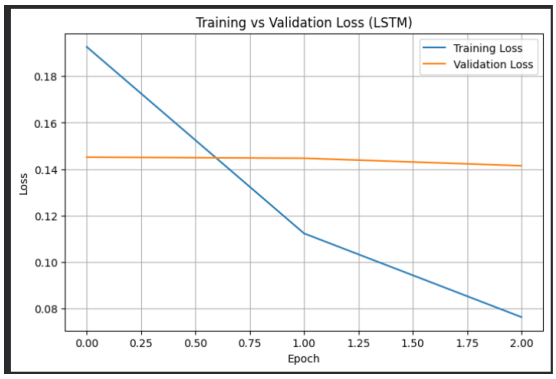


Fig. 11. Training Loss Graph

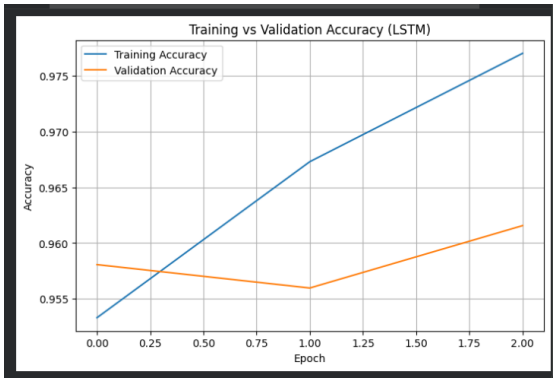


Fig. 12. Accuracy Graph

H. LSTM Model Result

From the table, the model perform good on the most of the real job class but recall is small class or fake job show the difficulty in detect the fake job posting.

I. LSTM confusion matrix

This matrix represent, True neg. 3384,True pos. 77, False neg. 104 and False pos. 11.

TABLE VI
PERFORMANCE OF LSTM MODEL ON FAKE JOB POSTING

Class	Precision	Recall	F1-score	Samples
Real Jobs (0)	0.97	1.00	0.98	3395
Fake Jobs (1)	0.88	0.43	0.57	181
Macro Average	0.92	0.71	0.78	3576
Weighted Average	0.97	0.97	0.96	3576

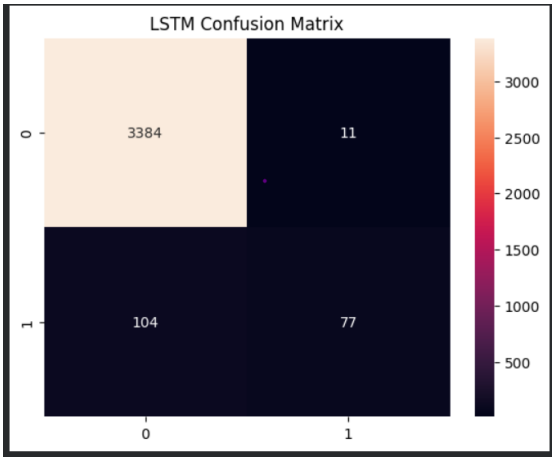


Fig. 13. LSTM Confusion Matrix

VI. FUTURE WORK AND CONCLUSION

The lack of class unbalanced control poor LSTM modeling and the lack of strong transformer model such as BERT restricted the paper potential for generalizations with its better result. Moreover, the use of train-test shows the more reliable performance predictions.

Future work should use the k-fold cross validation and pretrained embedded features.

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